`

USED CARS PRICE ANALYSIS

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INTRODUCTION

- Website Name: CARS24(CARS24 Services Private Limited)
- Industry: Indian Automotive Technology company

  Digital-first online marketplace for used vehicles
- Founded: 2015
- Headquarters: Gurugram, India
- CARS24's platform connects individual sellers, buyers, and dealers, enabling seamless transactions that remove many traditional frictions associated with the used-car market such as uncertain pricing, manual paperwork, and fragmented dealer networks. Its core offerings include AI-powered pricing and valuation tools, doorstep vehicle inspection, instant payment solutions, and complete paperwork management all designed to simplify the process for users.
- The company has rapidly transformed how used cars are bought, sold, financed, and managed by integrating technology, transparency, and end-to-end services across the automotive lifecycle.

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PROBLEM STATEMENT

- Car price variation based on various cars brands.
- Best cars with good quality with less price.

OBJECTIVES

- To collect used car listing data from the CARS24 platform through automated web scraping.
- To clean and preprocess the data by handling missing values, inconsistencies, and outliers.
- To analyze pricing trends and market patterns in the used car ecosystem.
- To identify key factors influencing used car prices such as brand, age, mileage, and fuel type.
- To derive actionable insights that support data-driven decision making.

METHODOLOGY

The methodology outlines the step-by-step process followed in the project, from data collection to final analysis and visualization. It ensures a structured approach to achieve the project's objectives by using various tools and techniques.

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STEP 1:

WEB SCRAPING

The goal of this phase is to scrape all the cities data from CARS24 website and store them .

Web Scraping Steps :

- Declaring the URL (ex : https://www.cars24.com/buy-used-cars-hyderabad/) .
- Automating the CARS24 website to get scrolled by using scrolling logic with selenium .
- Stored HTML content of the page .
- Converting messy HTML into well formatted and human readable structure using prettify() .
- Creating empty lists of required columns .
- Storing the data in empty lists .
- Converting the data into DataFrame .
- Saving the file .

Tools Used : Jupyter Notebook

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STEP 2:

DATA CLEANING & EDA

The goal of this phase is to ensure the dataset is free from errors, missing values, and irrelevant information.

Cleaning Steps :

- Checking for nulls (No null values present)
- Checking for duplicates (No Duplicates present)
- Dropping inessential columns .
- Standardizing the Column names .
- Standardizing the Data .
- Declaring current year and creating a new column car_age_years
  car_age_years = current year – manufacturing year
- Adding a new column city .
- Outliers Detection .
- Final Validation .
- Saving the file .

Tools Used : Jupyter Notebook

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STEP 3 :

SQL ANALYSIS

With a clean dataset, the next step is analysis using Sql, Through SQL, organizations can track performance, discover trends, understand customer behavior, and identify what drives profits or causes losses.

Key Metrics are :

OLDEST_MODEL_YEAR
2006

NEWEST_MODEL_YEAR
2025

LOWEST_PRICE
24000

HIGHEST_PRICE
13900000

TOTAL_BRANDS
31

TOTAL_CARS
18592

AVG_PRICE
526585.31

Analysis :

CITY	AVG_PRICE
Ludhiana	772731.09
Coimbatore	708032.79
Patna	653117.83
Hyderabad	644679.07
Lucknow	635463.11
Nashik	628716.13
ChandigarhTricity	614807.83
Bangalore	592724.92
Rajkot	573166.21
Nagpur	566311.98
Chennai	559944.54
Agra	559459.4
Pune	542889.95
Kochi	535450
Kolkata	518782.69
Mumbai	504172.19
Indore	499718.41
Surat	485919.05
Vadodara	472349.82
Jaipur	468942.03
Delhi NCR	438812.72
Ahmedabad	416317.27

CITY	NO_OF_CARS
Delhi NCR	4560
Bangalore	2392
Mumbai	1812
Hyderabad	1639
Pune	1472
Ahmedabad	1100
Chennai	613
Kolkata	520
Lucknow	488
Nagpur	484
Jaipur	483
Agra	431
Surat	420
Rajkot	367
Patna	314
Vadodara	283
ChandigarhTricity	281
Indore	277
Kochi	260
Nashik	155
Coimbatore	122
Ludhiana	119

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CAR_AGE_YEARS	AVG_PRICE
1	1114987.42
2	1098309.3
3	979710.31
4	870878.08
5	770056.98
6	686762.59
7	593852.84
8	492919.63
9	456105.53
10	394453.83
11	343855.59
12	300372.06
13	268627.07
14	224107.07
15	167221.14
16	167252.43
18	245000
19	74000
20	110000

KMS_RANGE	AVG_PRICE	MIN_PRICE	MAX_PRICE
0-20k	848402.41	33000	13900000
100k+	418153.66	39000	4011000
20k-40k	677592.38	28000	10500000
40k-60k	541924.98	31000	7799000
60k-80k	474290.29	24000	3564000
80k-100k	448786.49	44000	4200000

EMI	CARS
Low EMI(<8K)	5590
Mid EMI(8K-15K)	8335
High EMI(15K+)	4667

FUEL_TYPE	CARS	AVG_PRICE
Petrol	13333	466489.01
Hybrid	15	2130266.67
CNG	262	412816.79
Electric	124	1188338.71
Diesel	4858	675815.15

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BRAND	CAR_NAME	CITY	PRICE	CAR_AGE_YEARS	KMS_DRVEN	EMI
Tata	Tata PUNCH	Ahmedabad	455000	5	37733	8032
Tata	Tata Tiago	Chennai	455000	4	21680	8032
Nissan	Nissan MAGNITE	Delhi NCR	455000	4	36844	8032
Renault	Renault TRIBER	Surat	455000	4	24691	8036
Hyundai	Hyundai GRAND I10 NIOS	Vadodara	456000	4	25069	8046
Tata	Tata TIGOR	Mumbai	456000	5	36098	8050
Tata	Tata Tiago	Kochi	456000	3	29883	8050
Maruti	Maruti Celerio	Delhi NCR	456000	5	29007	8050
Tata	Tata PUNCH	Delhi NCR	456000	3	28002	8050
Renault	Renault TRIBER	Delhi NCR	457000	4	28210	8067
Tata	Tata ALTROZ	Delhi NCR	458000	4	33014	8085
Hyundai	Hyundai GRAND I10 NIOS	Jaipur	458000	5	17403	8085
Honda	Honda Amaze	Kolkata	460000	4	20831	8120
Tata	Tata ALTROZ	Mumbai	460000	4	10533	8120
Tata	Tata Tiago	Mumbai	460000	5	27304	8120
Maruti	Maruti Celerio	Mumbai	460000	3	10403	8120
Maruti	Maruti New Wagon-R	Mumbai	460000	5	54543	8120
Maruti	Maruti New Wagon-R	Patna	460000	4	22480	8120
Maruti	Maruti Alto K10	Rajkot	460000	2	40993	8120
Maruti	Maruti S PRESSO	Rajkot	461000	1	5300	8138
Maruti	Maruti New Wagon-R	Mumbai	461000	5	46069	8138
Renault	Renault TRIBER	Delhi NCR	461000	5	21553	8138
Maruti	Maruti New Wagon-R	Delhi NCR	462000	3	33092	8157
Tata	Tata Tiago	Chennai	462000	3	15012	8156
Maruti	Maruti New Wagon-R	Mumbai	462000	3	10329	8156
Tata	Tata ALTROZ	Mumbai	462000	5	41870	8156

TRANSMISSION_TYPE	CARS	AVG_PRICE
Manual	14506	437888.12
Auto	4086	841475.53

Tools Used : MS SQL

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STEP 4:

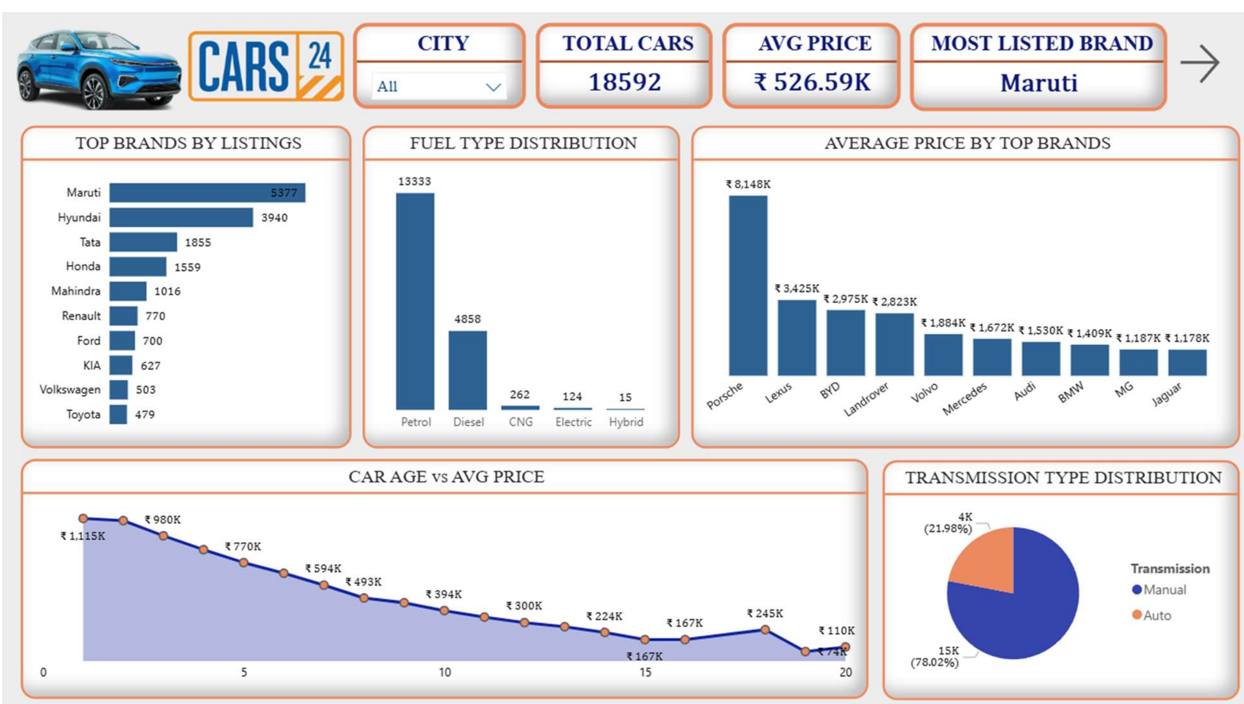
DATA VISUALIZATION

It helps people quickly understand patterns, trends, and relationships that might stay hidden in raw data.

Data visualization also makes communication easier, turning complex insights into simple stories that everyone in a business can understand and act on. In short, data visualization transforms data into a visual language that drives smarter, faster, and more confident decisions.

Tools Used: Power BI

Created Interactive Dashboards that includes :



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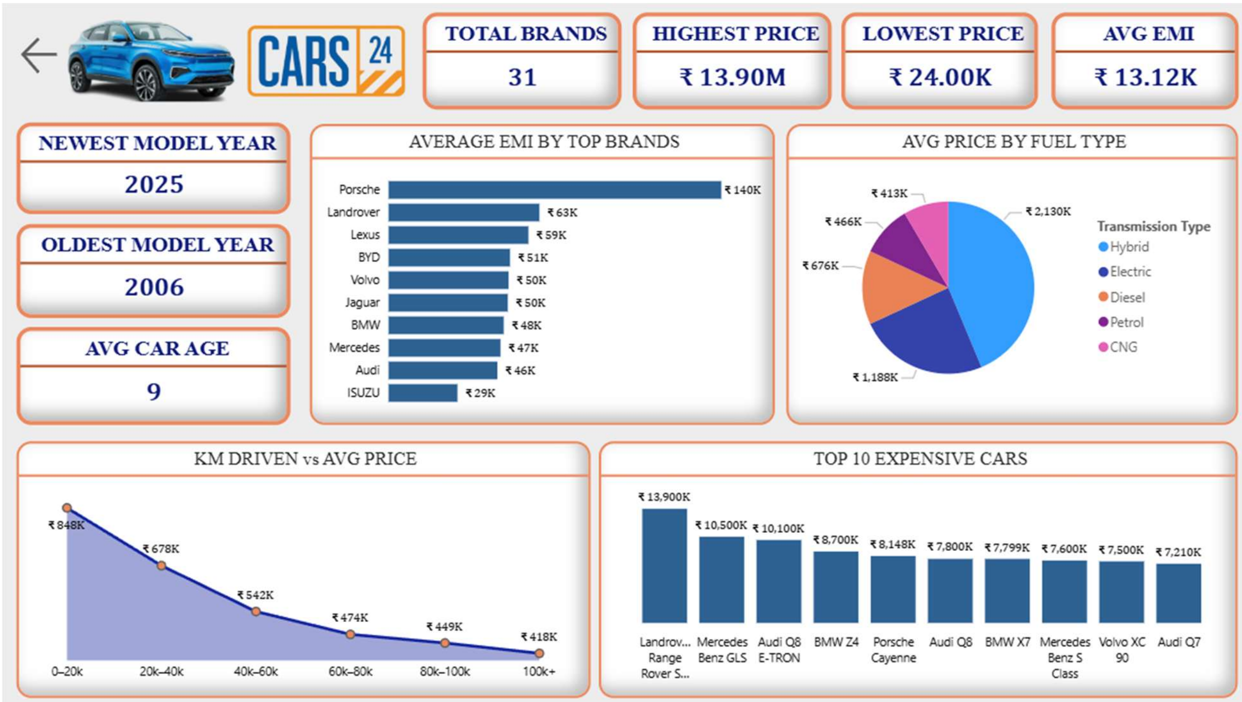
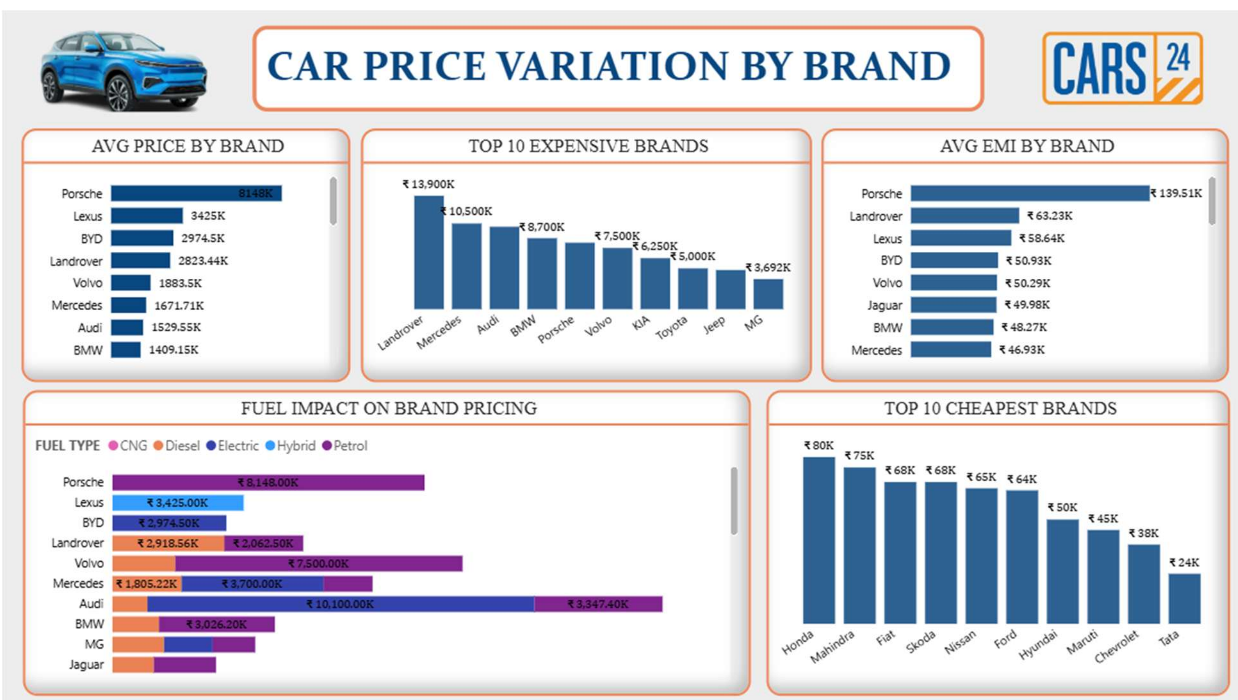

CARS 24
TOTAL BRANDS
31
HIGHEST PRICE
₹ 13.90M
LOWEST PRICE
₹ 24.00K
AVG EMI
₹ 13.12K
NEWEST MODEL YEAR
2025
OLDEST MODEL YEAR
2006
AVG CAR AGE
9
AVERAGE EMI BY TOP BRANDS
Porsche ₹140K
Landrover ₹63K
Lexus ₹59K
BYD ₹51K
Volvo ₹50K
Jaguar ₹50K
BMW ₹48K
Mercedes ₹47K
Audi ₹46K
ISUZU ₹29K
AVG PRICE BY FUEL TYPE
₹413K
₹2,130K
₹466K
₹676K
₹1,188K
Transmission Type
Hybrid
Electric
Diesel
Petrol
CNG
KM DRIVEN vs AVG PRICE
₹848K
₹678K
₹542K
₹474K
₹449K
₹418K
0-20k
20k-40k
40k-60k
60k-80k
80k-100k
100k+
TOP 10 EXPENSIVE CARS
₹13,900K
₹10,500K
₹10,100K
₹8,700K
₹8,148K
₹7,800K
₹7,799K
₹7,600K
₹7,500K
₹7,210K
Landrov... Range Rover S...
Mercedes Benz GLS
Audi Q8 E-TRON
BMW Z4
Porsche Cayenne
Audi Q8
BMW X7
Mercedes Benz S Class
Volvo XC 90
Audi Q7


CAR PRICE VARIATION BY BRAND
CARS 24
AVG PRICE BY BRAND
Porsche 8148K
Lexus 3425K
BYD 2974.5K
Landrover 2823.44K
Volvo 1883.5K
Mercedes 1671.71K
Audi 1529.55K
BMW 1409.15K
TOP 10 EXPENSIVE BRANDS
₹13,900K
₹10,500K
₹8,700K
₹7,500K
₹6,250K
₹5,000K
₹3,692K
Landrover
Mercedes
Audi
BMW
Porsche
Volvo
KIA
Toyota
Jeep
MG
AVG EMI BY BRAND
Porsche ₹139.51K
Landrover ₹63.23K
Lexus ₹58.64K
BYD ₹50.93K
Volvo ₹50.29K
Jaguar ₹49.98K
BMW ₹48.27K
Mercedes ₹46.93K
FUEL IMPACT ON BRAND PRICING
FUEL TYPE CNG Diesel Electric Hybrid Petrol
Porsche ₹8,148.00K
Lexus ₹3,425.00K
BYD ₹2,974.50K
Landrover ₹2,918.56K ₹2,062.50K
Volvo ₹7,500.00K
Mercedes ₹1,805.22K ₹3,700.00K
Audi ₹10,100.00K ₹3,347.40K
BMW ₹3,026.20K
MG
Jaguar
TOP 10 CHEAPEST BRANDS
₹80K
₹75K
₹68K
₹68K
₹65K
₹64K
₹50K
₹45K
₹38K
₹24K
Honda
Mahindra
Fiat
Skoda
Nissan
Ford
Hyundai
Maruti
Chevrolet
Tata

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CAR NAME	BRAND	PRICE	KMS_DRIVEN	CAR AGE YEARS	FUEL TYPE	TRANSMISSION TYPE	EMI	CITY
Tata PUNCH	Tata	455K	37733	5	Petrol	Manual	8032	Ahmedabad
Tata Tiago	Tata	455K	21680	4	Petrol	Manual	8032	Chennai
Nissan MAGNITE	Nissan	455K	36844	4	Petrol	Manual	8032	Delhi NCR
Renault TRIBER	Renault	455K	24691	4	Petrol	Manual	8036	Surat
Maruti Celerio	Maruti	456K	29007	5	Petrol	Auto	8050	Delhi NCR
Tata PUNCH	Tata	456K	28002	3	Petrol	Manual	8050	Delhi NCR
Tata Tiago	Tata	456K	29883	3	Petrol	Manual	8050	Kochi
Tata TIGOR	Tata	456K	36098	5	Petrol	Manual	8050	Mumbai
Hyundai GRAND I10 NIOS	Hyundai	456K	25069	4	Petrol	Manual	8046	Vadodara
Renault TRIBER	Renault	457K	28210	4	Petrol	Manual	8067	Delhi NCR
Tata ALTROZ	Tata	458K	33014	4	Petrol	Manual	8085	Delhi NCR
Hyundai GRAND I10 NIOS	Hyundai	458K	17403	5	Petrol	Manual	8085	Jaipur
Honda Amaze	Honda	460K	20831	4	Petrol	Manual	8120	Kolkata
Maruti Celerio	Maruti	460K	10403	3	Petrol	Manual	8120	Mumbai
Maruti New Wagon-R	Maruti	460K	54543	5	Petrol	Manual	8120	Mumbai
Tata ALTROZ	Tata	460K	10533	4	Petrol	Manual	8120	Mumbai
Tata Tiago	Tata	460K	27304	5	Petrol	Auto	8120	Mumbai
Maruti New Wagon-R	Maruti	460K	22480	4	Petrol	Manual	8120	Patna

KEY FINDINGS

- There are 18592 cars with 31 unique brands listed .
- Most listed brand is ‘ Maruti ‘ .
- Highest price is 13.9M and Lowest price is 24k .
- Average price is 526.59k.
- Landrover Range Rover Sport is the most expensive car with a price of 1.39 Cr and Tata Nano is the most cheapest car with 24k .
- As the car age and kms driven is increasing price is getting dropped .
- There are more cars with Manual Transmission than Automated Transmission .
- Expensive brands has more EMI than budget brands due to their price difference .

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KEY INSIGHTS

- Car price is strongly influenced by vehicle age and kilometers driven .
- Brand reputation plays a major role in price retention .
- Automatic transmission cars are priced higher compared to manual cars, reflecting increasing urban demand.
- Outliers in pricing are often linked to luxury brands, very low mileage, or rare model variants.

CHALLENGES FACED

- Dynamic website structure made data extraction challenging due to JavaScript-rendered content and frequent DOM changes.
- Anti-scraping mechanisms such as rate limiting and dynamic class names increased the complexity of reliable data collection.
- Inconsistent data formats across listings (price, mileage, year, location) required extensive cleaning and standardization.
- Duplicate listings across cities or reposted vehicles needed careful identification and removal.
- Outliers and extreme values in price and mileage distorted analysis and required domain-based filtering.

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- City-wise data imbalance, where some cities had significantly more listings than others, affecting comparative analysis.
- Limited access to backend data, restricting insights to only publicly available listing attributes.
- Time-consuming scraping process due to pagination, infinite scrolling, and delays to avoid blocking.

CONCLUSION

This project successfully demonstrated an end-to-end data analytics workflow by scraping, cleaning, and analyzing used car data from the CARS24 platform. Through systematic data preprocessing and exploratory data analysis, key factors influencing used car prices—such as vehicle age, brand, fuel type, transmission type,emi,km driven, and city—were identified. The analysis revealed noticeable pricing variations among cars due its brand value , vehicle age and also distance driven . Overall, the project provided meaningful insights into the used car market while showcasing practical skills in web scraping, data cleaning, data analysis, and visualization. These insights can support informed decision-making for buyers, sellers, and automotive marketplace stakeholders.

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REFERENCES

- **Data Sources** : CARS 24
  https://www.cars24.com/buy-used-cars/
- **Tools Used** : Python (Jupyter Notebook) , Power BI , MS SQL .